

K-MAPS

- We can minimize Boolean Expression of 2, 3, or 4 Variables Very easily using the k-map without using any Boolean algebra theorem. It uses graphical or tabular representation.
 - There are two forms of k-map according to the problem:
1) Sum of product (SOP) 2) product of sum (POS)
 - If we consider the combination of inputs for which the boolean function is '1', then we will get the boolean function, which is in standard "Sum of product" form after simplifying k-map.
or "Sum of minterm"
 - Similarly, if we consider the combination of inputs for which the Boolean function is '0' then we will get standard "Product of Sum" form after simplifying the k-map. ("Product of Maxterm")
- * Steps to Solve Expression using k-map:
- 1) Select k-map according to the number of variables.
 - 2) Identify minterms or maxterms as given in the problem.
 - 3) for SOP put 1's in blocks of k-map respective to minterm (0's elsewhere)
 - 4) for POS put 0's in blocks of k-map respective to the maxterm. (1's elsewhere).

→ Representation of K-Map.

1) 2-Variables

B	0	1
A	0	1
1	2	3

2) 3-Variables

BC	00	01	11	10
A	0	1	3	2
1	4	5	7	6

3) 4-Variables

AB \ CD	CD			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

→ finding out minterm & Maxterm for given Boolean Exp.

Eg

$$A'B + AB'C + ABC \quad [SOP \Rightarrow \text{minterm}] \Rightarrow 1$$

$$A'B(C+C') + AB'C + ABC$$

$$A'BC + A'BC' + AB'C + ABC$$

$$\begin{array}{cccc} 011 & 010 & 101 & 111 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 5 & 7 \end{array}$$

Minterms: (3, 2, 5, 7)

Eg

$$(A+C')(B+C)(A'+B'+C) \quad [POS \Rightarrow \text{Maxterms}] \Rightarrow 0.$$

$$[A+(B \cdot B') + C'] [A \cdot A' + B + C] [A' + B' + C]$$

$$(A+B+C')(A+B'+C)(A+B+C)(A'+B+C)(A'+B'+C)$$

$$\begin{array}{ccccc} 110 & 100 & 111 & 011 & 001 \\ 6 & 4 & 7 & 3 & 1 \end{array}$$

Maxterm: (6, 4, 7, 3, 1)

~~KMAP~~

* if you want to find out Minterms, through the Maxterms, then simply put 1's on the values except the values/blocks present in (blocks) maxterms.

We can say that the terms except the Maxterms are minterms. Similarly the terms except the minterms are Maxterms.

$$\text{Eg: } \textcircled{1} \text{ Maxterm} \Rightarrow (6, 4, 7, 3, 1) \Rightarrow \text{corresponding Minterm} (2, 5, 0)$$

$$\textcircled{2} \text{ Minterm} \Rightarrow (3, 2, 5, 7) \Rightarrow \text{corresponding Maxterms} (0, 1, 4, 6)$$

K-Map Questions -

* Rules for K-Map

- ① No zeros allowed [We solve K-map only using 1's]
- ② No diagonals
- ③ Only power of 2 number of cells in each group
- ④ Groups should be as large as possible
- ⑤ Every 'one' must be in at least one group
- ⑥ Overlapping allowed.
- ⑦ Wrap around allowed
- ⑧ Get the fewest number of groups possible.

Eg ① 2-Variable K-Map

$$F(A, B) = \sum (m_0, m_1, m_2) = \bar{A}\bar{B} + \bar{A}B + A\bar{B}$$

00 01 10

(0, 1, 2) → Minterms.

	B	0	1
A	0	1	1
	1	1	0

from group 1 → $\bar{A}$

from group 2 → $\bar{B}$

Simplified Boolean Exp. → $\bar{A} + \bar{B}$

Eg ② 3-Variable K-Map.

$$F(A, B, C) = \sum (m_0, m_1, m_2, m_4, m_5, m_6)$$

so, Minterms will be (0, 1, 2, 4, 5, 6)

	BC	00	01	11	10
A	0	1	1	0	1
	1	1	1	1	0

from g1 → $\bar{B}$

from g2 → $\bar{C}$

Simplified Boolean Exp. → $\bar{B} + \bar{C}$

Ex 3

4-Variable k-map

$$F(A, B, C, D) = \sum(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$$

So, minterms will be $\Rightarrow (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$

CD \ AB	00	01	11	10
00	1	0	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

from $g_1 \rightarrow C'$
from $g_2 \rightarrow A'D'$
from $g_3 \rightarrow BD'$

Simplified Boolean Exp $\rightarrow C' + A'D' + BD'$

$\Rightarrow$ Don't Care condition in k-map $\rightarrow$ The Don't Care conditions allow us to replace the empty cell of a k-map to form a grouping of the variable which is larger than that of forming groups without don't care.

• while forming groups of cells we can consider a don't care cell as 1 or 0 or we can also ignore that cell. Therefore the don't care condition can help us to form a larger group of cells.

• Don't Care can be represented as $\rightarrow X, -, \phi$

Ex $\rightarrow F = m(1, 5, 6, 11, 12, 13, 14) + d(4)$

CD \ AB	00	01	11	10
00	0	1	3	2
01	X	1	7	6
11	1	1	15	14
10	8	9	11	10

from $g_1 \rightarrow A'C'D$

from $g_2 \rightarrow BC'$

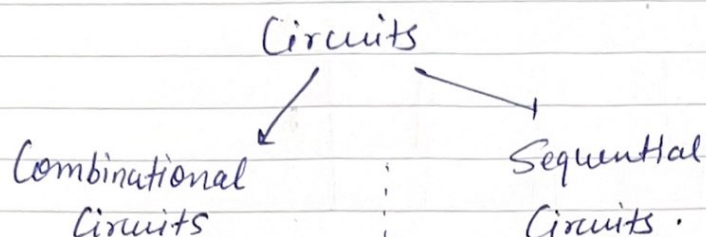
from $g_3 \rightarrow BD'$

from $g_4 \rightarrow AB'CD$

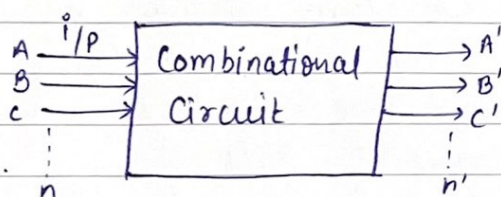
Simplified Exp

$$BC' + BD' + A'C'D + AB'CD$$

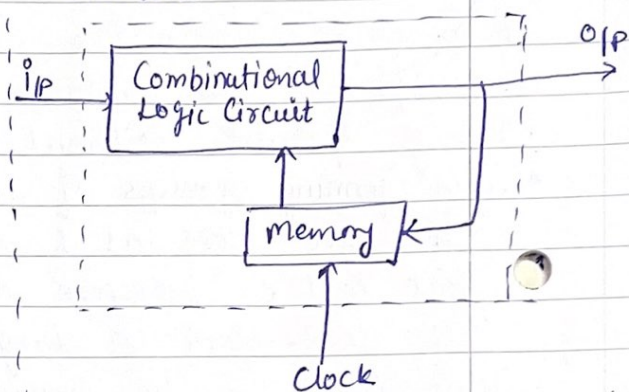
⇒ Combinational Circuits: A circuit is a model of computation in which i/p values proceed through a sequence of gates, each of which computes a function.



- Combinational circuit is a circuit in which we combine different gates in the circuit.
- It doesn't contain any memory element so the o/p of combi. circuits at any instant of time, depends only on the levels present at i/p terminal.



- Sequential circuits contain memory, so o/p can vary based on i/p. This type of circuit uses previous i/p, o/p, clock and a memory element.

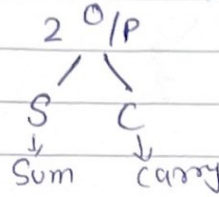
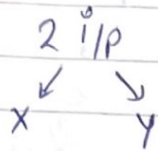


Combinational Circuits

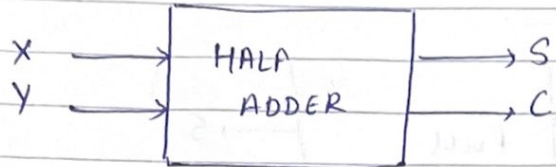
- 1) Half Adder
- 2) Full Adder
- 3) Adder Subtractor
- 4) Multiplexer / demultiplexer

5) Decoders.

1) HALF ADDER!



a) Block diagram



b) TRUTH Table

X	Y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$\begin{cases} \bar{x}y \\ xy \end{cases} \rightarrow \text{minterms for Sum}$

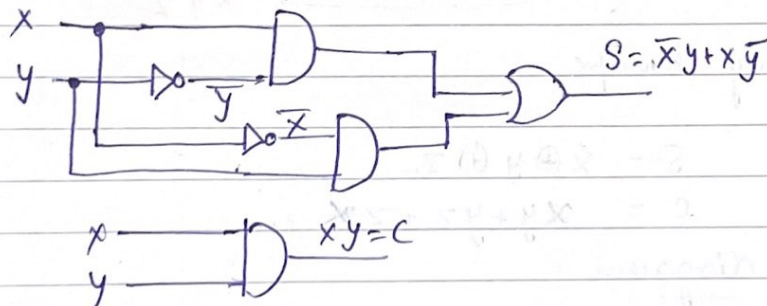
$\{xy \rightarrow \text{minterm for Carry}$

c) Circuit diagram:

Using the minterms

$$\text{Sum} \rightarrow S = \bar{x}y + x\bar{y}$$

$$\text{Carry} \rightarrow C = xy$$

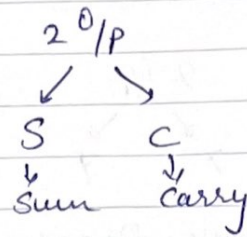
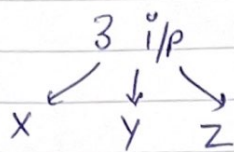


~~altern~~ $\bar{x}y + x\bar{y}$ can also known as X-OR

so $S = x \oplus y$



Full Adder:



a) Block diagram



b) Truth Table:

x	y	z	C	S	minterm
0	0	0	0	0	Sum
0	0	1	0	1	$\bar{x}\bar{y}z$
0	1	0	0	1	$\bar{x}y\bar{z}$
0	1	1	1	0	$\bar{x}yz$ minterm for carry
1	0	0	0	1	$x\bar{y}\bar{z}$
1	0	1	1	0	$x\bar{y}z$
1	1	0	1	0	$xy\bar{z}$
1	1	1	1	1	xyz

$\rightarrow x y z$

or using kmaps

$$S = x \oplus y \oplus z$$

$$C = xy + yz + zx$$

c) Circuit diagram: